

Hiver SDE Intern — Take-Home Assignment

AI SUPPORT AGENT FOR AMAZONHELP

Evaluation Report

GitHub Repository:

<https://github.com/nithimegha/hiver-support-agent>

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Executive Summary

This project evaluates a prototype AI support agent for AmazonHelp. For each customer message, the system predicts a support intent, retrieves a similar historical AmazonHelp case, uses the historical resolution as grounding for a draft reply, and recommends AUTO-HANDLE or ESCALATE.

What good means

- Correctly identify the customer's primary support need.
- Retrieve genuinely relevant historical evidence.
- Produce a useful response grounded in that evidence.
- Automate only when evidence is strong and escalate uncertain or sensitive cases.

Headline Results

Evaluation	Accuracy	Macro F1
Majority-class baseline	69.54%	0.1025
TF-IDF + Logistic Regression	93.15%	0.8711
Golden set — 152 examples	17.76%	0.1144

Key takeaway

The classifier is strong against the existing dataset-derived labels, but the independent golden-set result shows that the 93.15% headline should not be interpreted as production reliability. Taxonomy quality, evidence retrieval, and independent evaluation are the main areas requiring further work.

1. Problem Framing and System Design

Scope

The prototype focuses on the core support-agent decision loop rather than production infrastructure.

- No live Amazon order/account APIs or customer authentication.
- No live Twitter/X integration.
- No automatic refunds, cancellations, account changes, or other transactions.
- No production deployment, security/monitoring stack, multilingual support, or model finetuning.

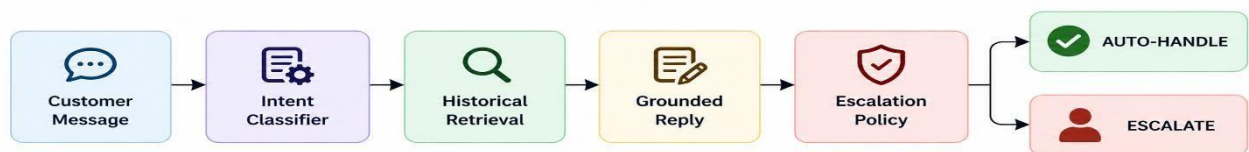
Dataset and Brand Selection

- Source: Customer Support on Twitter (thoughtvector/customer-support-on-twitter).
- Approximately 3M tweets across multiple brands, including multi-turn support conversations.
- AmazonHelp was selected because it has a large volume of customer-support interactions.
- Direct AmazonHelp customer-response pairs were extracted for historical evidence.
- 20,000 historical AmazonHelp cases were indexed for retrieval.

Intent Taxonomy

- order_delivery
- refund_return
- payment_billing
- account_access
- cancellation_subscription
- product_device_issue
- digital_service_issue
- seller_marketplace

Architecture



The policy escalates when retrieval similarity is below the configured threshold or when the predicted intent is account access or payment billing. Otherwise, sufficiently supported cases are recommended for AUTO-HANDLE.

2. Evaluation Method and Results

Classification Baselines

Approach	Type	Accuracy	Macro F1
Majority class	Trivial	69.54%	0.1025
TF-IDF + Logistic Regression	Simple ML	93.15%	0.8711

The majority baseline always predicts order delivery, the most frequent class. The TF-IDF + Logistic Regression model substantially improves over this trivial baseline on the held-out dataset.

Independent Golden Evaluation

- 152 examples were manually labelled using the eight-intent taxonomy.
- Examples include clear requests and ambiguous support messages.
- The set provides an independent check rather than relying only on dataset-derived labels.
- Golden-set performance: 17.76% accuracy and 0.1144 macro F1.

Reply Quality Evaluation

A local Qwen2.5-0.5B-Instruct model was used as an LLM-as-judge. Each reply is scored from 1–5 for correctness, groundedness, relevance, helpfulness, tone, and overall quality.

Human vs LLM judge	Result
Exact agreement on overall score	4/8 (50%)
Agreement within ± 1 point	6/8 (75%)

The overlap is small, so the judge is treated as supporting evidence rather than ground truth.

Escalation Policy Check

The escalation policy was tested on 8 policy cases and produced 8/8 expected outcomes. This is policy/unit-test consistency, not evidence of real-world escalation accuracy.

3. Failure Analysis

Confusion between related intents

Observed example: “Why did my order get declined?”

Hypothesis: The taxonomy compresses support issues that can overlap.

Improvement: Define clearer boundaries and use conversation context.

Context-poor messages

Observed example: “Still not open packet.bcoz amazon gav me optn to return by 11nov.”

Hypothesis: A single tweet can omit information needed for classification.

Improvement: Use the full conversation thread where available.

Retrieval mismatch

Observed example: “My order is late. Can you check the delivery?”

Hypothesis: Semantic similarity does not guarantee the same underlying resolution.

Improvement: Filter by intent, retrieve multiple candidates, and add evidence checks.

Customer-specific historical wording

Observed example: Historical replies may mention a particular courier, delivery time, order, or customer action.

Hypothesis: Copying a historical reply can transfer details that do not apply to the new customer.

Improvement: Generate a fresh response from the historical evidence.

Taxonomy/label ambiguity

Observed example: Some messages combine delivery, payment, account, or order-management signals.

Hypothesis: A compact single-label taxonomy can be difficult to apply consistently.

Improvement: Add inclusion/exclusion rules and perform a second annotation pass.

1. What Is Misleading About My Headline Number?

93.15% accuracy is not overall agent reliability

The most misleading number is the 93.15% classification accuracy. It is measured against the labels used for the held-out dataset, not against an independently adjudicated support taxonomy.

- The golden set of 152 manually labelled examples gives 17.76% accuracy and 0.1144 macro F1.
- Manual review exposed difficult intent boundaries and label/taxonomy uncertainty.
- Therefore, strong performance against dataset labels does not establish reliable humanlevel routing.
- The system should not be deployed for fully autonomous support based on the 93.15% figure alone.

The correct interpretation

The simple classifier performs strongly against the existing dataset labels, while independent human evaluation reveals substantial risks around taxonomy quality, intent ambiguity, and generalization.

Other limitations

- Retrieval uses 20,000 historical cases rather than the complete support history.
- Historical responses are used directly as grounded drafts, so case-specific wording can remain.
- The local LLM judge is a small model and was compared with humans on only eight overlapping examples.
- The escalation evaluation is a policy/unit test, not a production outcome study.

Why this matters

The evaluation gap is itself an actionable product finding: improving annotation quality and taxonomy clarity is as important as improving the classifier.

5. One-Week Improvement Plan

Days 1–2: Taxonomy and labels

- Review golden-set disagreements.
- Define inclusion/exclusion rules for every intent.
- Adjudicate ambiguous examples.
- Expand the golden set toward 250 examples.

Day 3: Retrieval

- Use predicted intent as a retrieval filter.
- Retrieve multiple historical candidates.
- Add evidence-quality checks.
- Remove customer-specific details before response use.

Day 4: Response generation

- Generate a fresh response from retrieved evidence.
- Block unsupported claims.
- Keep evidence separate from generated wording.

Day 5: Evaluation

- Increase the human/LLM judge overlap set.
- Calibrate the LLM judge against human ratings.
- Add targeted tests for difficult intent boundaries.
- Measure retrieval quality separately from response quality.

Days 6–7: Safety and productization

- Combine classification confidence and retrieval quality for escalation.
- Expand human review for uncertain cases.
- Add structured decision logging.
- Test on unseen conversations before considering production deployment.

Key Engineering Decisions

- Selected AmazonHelp for high interaction volume.
- Used direct customer-response pairs as historical evidence.
- Kept the taxonomy to eight understandable intents.
- Used majority and TF-IDF baselines as trivial and simple reference points.
- Used sentence embeddings and FAISS for semantic retrieval.
- Added similarity-based escalation and human review for sensitive intents.
- Used an independent golden set and human/LLM comparison to avoid relying on a single automated metric.

Conclusion

The AmazonHelp AI Support Agent demonstrates a practical approach to automating customersupport workflows by combining intent classification, historical-case retrieval, grounded reply generation, and rule-based escalation. The system achieved strong performance against the heldout test split, while the independent golden-set evaluation revealed important gaps in generalization and label consistency. This highlights that headline model accuracy alone is not sufficient to measure real-world support quality. Overall, the project provides a reproducible baseline and a clear foundation for improving classification reliability, retrieval quality, reply helpfulness, and safe human escalation in future iterations.